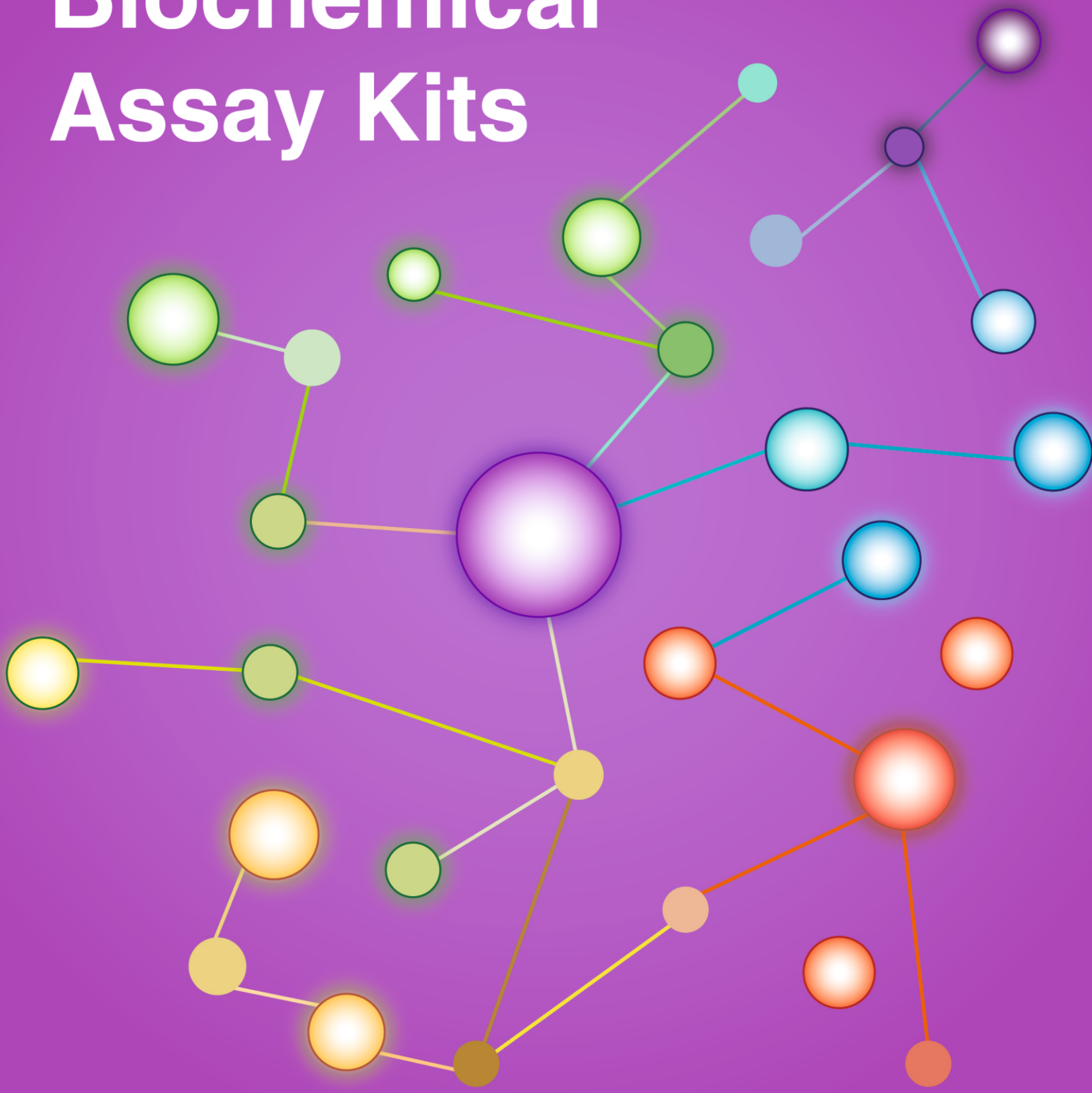
Solarbio®
LIFE SCIENCES

Sri Saradi Scientific

Biochemical Assay Kits



Beijing Solarbio Science & Technology Co.,Ltd.

Introduction

Solarbio biochemical assay kits are developed on the basis of the principle involved chemical reaction/enzymatic reaction or material structure characteristics which can be used to detect the content of substances or enzyme activity in samples.Solarbio has independently developed 25 series which more than 500 varieties of biochemical assay kits.Our kits are ready-to-use products which characterized by simple,rapid,stability and batch testing. Our kits covered many fields including agriculture,forestry,medicine, pharmacy, food, environmental protection and so on. These kits are applied for detecting sample from soil, water quality, animal/plant tissues, serum, cells, bacteria to food, etc.Detection targets is including oxidative stress assay,co-enzymes&cofactors metabolism,fatty acid&lipid metabolism, carbohydrate metabolism,inorganic ions metabolism, soil system metabolism, nitrogen metabolism, energy metabolism, protease activity assay and so on.We also provide customized products and detection service for biochemical assay kits.

The catalogue number of biochemical assay kits are coded ending with "0" or "5", representing two types of reaction systems respectively. The catalogue number ending with "0" represents spectrophotometer (1 mL of the reaction system) which can be detected by spectrophotometry.The catalogue number ending with "5" represents micro-determination(0.2 mL of the reaction system) which can be detected by spectrophotometer(Visible range/UV)or microplate reader.

Product Catagories

1. Oxidative Stress
2. Soil System Metabolism
3. Carbohydrate Metabolism
4. Lipid Metabolism
5. Energy Metabolism
6. Protease Activity Assay
7. Nitrogen Metabolism
8. Coenzymes&Cofactors Metabolism
9. Inorganic Ions Assay

Oxidative Stress

Vitamin C metabolism

- Ascorbic Acid(AsA)
- Dehydroascorbic Acid(DHA)
- L-galactose-1,4-lactone dehydrogenase(Gal LDH)
- Ascorbic Acid Oxidase(AAO)
- Ascorbate Peroxidase(APX)
- Monodehydroascorbate Reductase(MDHAR)
- Dehydroascorbate Reductase(DHAR)

Oxidant assay

- Hydrogen Peroxide(H_2O_2)
- Malondialdehyde(MDA)
- Xanthine Oxidase(XOD)
- Glucose Oxidase(GOD)
- Polyphenol Oxidase(PPO)
- Protein Carbonyl
- Diamine Oxidase(DAO)
- Superoxide Anion

Glutathione assay

- Reduced Glutathione(GSH)
- Oxidized Glutathione(GSSG)
- Glutathione Peroxidase(GPX)
- Thioredoxin Peroxidase(TPX)
- Glutathione S-transferase(GST)
- Glutathion Reductases(GR)
- Oxidized Thioredoxin Reductase(TrxR)
- γ-glutamylcysteine Ligase(GCL)
- γ-glutamyl Transpeptidase(γ-GT)

Antioxidant assay

- Superoxide Dismutase(SOD)
- Catalase(CAT)
- Peroxidase(POD)
- Ceruloplasmin(CP) Content A
- Total Antioxidant Capacity(T-AOC)
- Hydroxyl Free Radical Scavenging Capacity
- Thioredoxin Peroxidase(TPX)
- Glutathione Peroxidase(GPX)
- Glutathione S-transferase(GST)
- Ascorbic Acid(AsA)
- Total Mercapto(-SH)
- Thiol (Non-Protein Sample)
- Tannic Acid
- Pectin
- Cinnamic acid 4-hydroxylase(C4H)
- Tyrosinase
- Anthocyanidin Reductase
- Indoleacetic acid oxidase
- Plant Anthocyanin/Flavonoids/Proanthocyanidins
- Uric Acid(UA)

Peroxidation assay

- Superoxide Dismutase(SOD)
- Catalase(CAT)
- Peroxidase(POD)
- Polyphenol Oxidase(PPO)
- Phenylalnine Ammonialyase (PAL)
- Proline(PRO)

Soil System Metabolism

Soil system metabolism

- Soil Urease(S-UE)
- Soil Polyphenoloxidase
- Soil β-glucosidase(β-GC)
- Soil Cellulase(S-CL)
- Soil Catalase(S-CAT)
- Soil Nitrate Reductase(S-NR)
- Soil Saccharase(S-SC)
- Soil acid phosphatase(S-ACP)
- Soil Neutral Phosphatase(S-NP)
- Soil Alkaline Phosphatase(S-AKP/ALP)
- Soil Hydrargyrum(S-Hg)
- Soil Phosphate(S-PHOS)
- Soil Phosphorus
- Soil Dehydrogenase
- Soil Acid Protease
- Soil Neutral Protease
- Soil Alkaline Protease
- Soil FDA Hydrolase
- Soil Nitrate Reductase(S-NR)
- Soil Peroxidase
- Soil Diastase
- Soil Neutral Invertase(S-NI)
- Soil Acid Invertase(S-NI)
- Soil α-glucosidase(S-α-GC)
- Soil Lipase(S-LPS)
- Soil Aryl Sulfatase
- Soil N-Acetyl-β-D-glucosaminidase(S-NAG)
- Soil Laccase
- Soil Leucine Arylamidase(S-LAP)

Carbohydrate Metabolism

Glycolysis
Hexokinase(HK)
Pyruvate Kinase(PK)
Phosphofructokinase(PFK)
Phosphoenolpyruvate Carboxylase(PEPC)
Pyruvate(PA)
Lacate Dehydrogenase(LDH)
Glyceraldehyde-3-phosphate Dehydrogenase(GAPDH)

Tricarboxylic acid cycle
αKetoglutarate Dehydrogenase(α-KGDH)
Citric Acid(CA)
Succinate Dehydrogenase(SDH)
Pyruvate Dehydrogenase(PDH)
Isocitrate Dehydrogenase Mitochondrial(ICDHm)
Citrate Synthase(CS)
Acetyl Co-A
Mitochondrial Citric Acid(MCA)

Sucrose metabolism
Sucrose Synthetase(SS)
Neutral Invertase(NI)
Acid Invertase(AI)
Plant Saccharase
Sucrose Phosphoric Acid Synthetase(SPS)
Sucrose Phosphorylase

Saccharides metabolism
Glucose
Glucogen
Trehalose
Sorbitol
Reducing Sugar(RS)
Blood Glucose
Starch
Tissue Glucose
Plant Soluble Sugar/Tissue Fructose
Plant Sucrose
Total Carbohydrate
Neutral Xylanase
Acidic Xylanase
Alkaline Xylanase
Cellulase(CL)
Trehalase
Sorbitol Dehydrogenase(SDH)
β-1,3-glucanase(β-1,3-GA)
α-glucosidase(α-GC)
β-glucosidase(β-GC)
α-galactosidase(α-GAL)
β-galactosidase(β-GAL)
Chitinase
β-xylosidase
ADPG Pyrophosphorylase(AGP)
α-amylase/β-amylase
Soluble Starch Synthase(SSS)
Bound Station Amylosynthease
Starch Branching Enzyme(SBE)

Lipid Metabolism

Fatty acid metabolism
Nonesterified Free Fatty Acids(NEFA)
Fatty Acid Synthetase
Fatty Acid Synthetase
Lipase(LPS)
Alcohol Dehydrogenase(ADH)
Acyltransferase(AAT)
Pyruvate Decarboxylase(PDC)
Triglyceride(TG)
Total Cholestenone(TC)
Free Cholestenone(FC)
Plant Lipoxygenase(LOX)

Lipase activity assay
Acetylcholinesterase(AchE)
Acid Phosphatase(ACP)
Alkaline Phosphatase(AKP/ALP)
Carboxylesterase(CarE)

Nitrogen Metabolism

Amino acid metabolism
Glutamic-pyruvic Transaminase(GPT)
Glutamic-oxalacetic Transaminase(GOT)
Glutamine Synthetase(GS)
Proline(PRO)
Amino Acid(AA)
Cysteine(Cys)
Glutamic Acid(Glu)
Hydroxyproline(HYP)

Nitrogen metabolism
Nitrate Reductase(NR)
Glutamate Synthase(GOGAT)
Glutaminase(GLS)
Glutamic Acid Dehydrogenase(GDH)
Nitric Oxide(NO)
Nitrite-Water and Soil sample
Nitrite-Food sample
Glutamine Synthetase(GS)

Energy Metabolism

ATP metabolism
ATP
Na^+K^+——ATPase
$Ca^{2+}Mg^{2+}$——ATPase

Electron transport chain
Transhydrogenase-1
Transhydrogenase-2
Electron Transport Chain Complex Ⅱ
Electron Transport Chain Complex Ⅲ
Electron Transport Chain Complex Ⅳ
Electron Transport Chain Complex Ⅴ

Cytochrome P450
NADPH-cytochrome C Reductase
Aminopyrine-N-demethylase
Aniline-4 hydroxylase(AH)
Erythromycin-N-methylenzyme
Cytochrome b5

Protease Activity Assay

Protease activity assay
Acidic Proteinase
Neutral Proteinase
Alkali Proteinase
Trypsase
Pepsase
Chymotrypsin

Coenzymes & Cofactors Metabolism

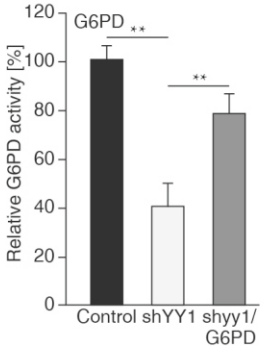
Coenzymes I metabolism
Coenzyme Ⅰ NAD(H)
NAD Kinase(NADK)
Lacate Dehydrogenase(LDH)
NAD-Malate Dehydrogenase(NAD-MDH)
NADP-Malate Dehydrogenase(NAD-MDH)
NADH Oxidase(NOX)
Citrate Synthase(CS)
Pyruvate Decarboxylase(PDC)
Alcohol Dehydrogenase(ADH)
Monodehydroascorbate Reductase(MDHAR)
Acetaldehyde Dehydrogenase(ALDH)

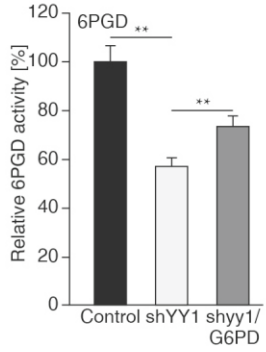
Coenzymes II metabolism
Coenzyme Ⅱ NADP(H)
NAD Kinase(NADK)
NADP Phosphatase(NADPase)

Inorganic Ions Assay

Inorganic ions metabolism
Phosphate/Total Phosphorus
Blood Calcium/Potassium/Magnesium/Sodium
Blood Phosphate/Zinc
Serum Ferri Ion
Serum Total Iron Binding Capacity(TIBC)
Water Mercury Ion(Hg^{2+})
Water Chromium(VI)
Soil Hydrargyrum(S-Hg)/Phosphorus
Soil Phosphate(S-PHOS)







Representative Papers

Journal	IF	Product	PMID
Signal Transduction and Targeted Therapy	40.8	ATP Content Assay Kit(BC0300)	38782919
Circulation	35.5	Mitochondrial complex I/Mitochondrial complex II/Mitochondrial complex III/Mitochondrial Complex IV/Mitochondrial Complex V Activity Assay Kit(BC0515/BC3235/BC3245/BC0945/BC1445)	38362779
Molecular Cancer	27.7	Lactic Acid(LA) Content Assay Kit(BC2235)	38711083
Cell Metabolism	27.7	Mitochondrial complex I/Mitochondrial complex II/Mitochondrial complex III/Mitochondrial Complex IV/Mitochondrial Complex V Activity Assay Kit(BC0510/BC3230/BC3240/BC0940/BC1440)	38232735
Cell Metabolism	27.7	Hexokinase(HK) Activity Assay Kit(BC0745)	39471815
Cell metabolism	27.7	Ketone Body Content Assay Kit(BC5065)	39142286
Advanced Materials	27.4	Reduced Glutathione(GSH) Content Assay Kit/Oxidized Glutathione(GSSG) Content Assay Kit/Malondialdehyde(MDA) Content Assay Kit(BC1175/BC1180/BC0025)	38241751
Advanced Materials	27.4	Citric Acid(CA) Content Assay Kit/α-Ketoglutarate Dehydrogenase(α-KGDH) Activity Assay Kit/ Succinate Dehydrogenase(SDH) Activity Assay Kit/NADP-Malate Dehydrogenase(NAD-MDH) Activity Assay Kit(BC2155/BC0715/BC0955/BC1055)	38295393
Advanced Materials	27.4	Hydroxyproline(HYP) Content Assay Kit(BC0250)	36862807
Advanced Materials	27.4	Lactic Acid(LA) HPLC Assay kit(BC2234)	39775989
Advanced Materials	27.4	Malondialdehyde(MDA) Content Assay Kit(BC0025)	39600046
Advanced Materials	27.4	Ascorbic Acid/Total Ascorbic Acid(AsA/T-AsA) Assay Kit (Colorimetric)(BC4630)	39410721
Advanced Materials	27.4	Hexokinase(HK) Activity Assay Kit(BC0745)	38469918
Advanced Materials	27.4	H_2S content Assay Kit(BC2055)	38943427
Advanced Materials	27.4	Reduced Glutathione(GSH) Content Assay Kit(BC1170)	39022876
Advanced Materials	27.4	Glycosylated Serum Protein Assay Kit(NBT-Colorimetric Method)(BC4940)	38308338
Advanced Materials	27.4	Coenzyme II NADP(H) Content Assay Kit/Reduced Glutathione(GSH) Content Assay Kit/ Oxidized Glutathione(GSSG) Content Assay Kit(BC1100/BC1175/BC1180)	39468892
iMeta	23.7	Nithine-δ-aminotransferase(δ-OAT) Activity Assay Kit/Lactic Acid(LA) Content Assay Kit/ Pyruvate(PA) Content Assay Kit(BC4405/BC2235/BC2205)	38882489
Cell Host & Microbe	20.6	Micro Hydroxyl Free Radical Scavenging Capacity Assay Kit/Superoxide Anion Activity Content Assay Kit/Reduced Glutathione(GSH) Content Assay Kit/Pyruvate Decarboxylase (PDC) Activity Assay Kit/DPPH Free Radical Scavenging Capacity Assay Kit (BC1325/BC1295/BC1175/BC1075/BC4755)	39197456
Bioactive Materials	18	Hydroxyproline(HYP) Content Assay Kit(BC0250)	39906648
Bioactive Materials	18	Reduced Glutathione(GSH) Content Assay Kit/Oxidized Glutathione(GSSG) Content Assay Kit(BC1175/BC1180)	39559423
Molecular Plant	17.1	Malondialdehyde(MDA) Content Assay Kit/Superoxide Dismutase(SOD) Activity Assay Kit(BC0025/BC0175)	38327054
National Science Review	16.3	Reduced Glutathione(GSH) Content Assay Kit(BC1170)	38410827
Angewandte Chemie-International Edition	16.1	Reduced Glutathione(GSH) Content Assay Kit(BC1175)	39325096
ACS Nano	15.8	Lacate Dehydrogenase(LDH) Activity Assay Kit(BC0685)	39806273
ACS Nano	15.8	ABTS Free Radical Scavenging Capacity Assay Kit/DPPH Free Radical Scavenging Capacity Assay Kit(BC4770/BC4750)	39102149
Drug Resistance Updates	15.8	Malondialdehyde(MDA) Content Assay Kit(BC0025)	39657434
ACS Nano	15.8	β-Hydroxybutyric Acid(β-HB) Content Assay Kit(BC5080)	39639502
Journal of Extracellular Vesicles	15.5	Lacate Dehydrogenase(LDH) Activity Assay Kit(BC0685)	39711510
Nature Communications	14.7	Nitrate Reductase(NR) Activity Assay Kit(BC0080)	39753568
Nature Communications	14.7	Nitrate Reductase(NR) Activity Assay Kit(BC0080)	39915505
Nature Communications	14.7	Cellulose Content Assay Kit(BC4280)	39578470
Nature Communications	14.7	Malondialdehyde(MDA) Content Assay Kit/Pyruvate Dehydrogenase(PDH) Activity Assay Kit(BC0025/BC0385)	39424837
Journal of the American Chemical Society	14.4	Reduced Glutathione(GSH) Content Assay Kit(BC1175)	39073804
Advanced Science	14.3	Malondialdehyde(MDA) Content Assay Kit(BC0025)	39804950
Advanced Science	14.3	Superoxide Dismutase(SOD) Activity Assay Kit/Catalase(CAT) Activity Assay Kit/ Peroxidase(POD) Activity Assay Kit/Malondialdehyde(MDA) Content Assay Kit (BC0170/BC0205/BC0090/BC0025)	39364742
Advanced Science	14.3	ABTS Free Radical Scavenging Capacity Assay Kit/DPPH Free Radical Scavenging Capacity Assay Kit(BC4770/BC4750)	39556697
Advanced Science	14.3	Reduced Glutathione(GSH) Content Assay Kit/Malondialdehyde(MDA) Content Assay Kit(BC1175/BC0025)	39630101
Biomaterials	12.8	Mitochondrial complex III/CoQ-cytochrome C reductase Activity Assay Kit/Mitochondrial Complex IV/Cytochrome C Oxidase Activity Assay Kit(BC3240/BC0940)	39467398

Sri Saradi Scientific

Authorised Exclusive Partner of Solarbio in India.

Beijing Solarbio Science & Technology Co.,Ltd.

301, Third Floor, Building 26, No. 16, Huanke Middle Road, Beijing Economic-Technological Development Area, Beijing, China

info@solarbio.com http://www.solarbio.com